

**Control: only the recognition residue (L122R) shifts the motif;
mutating unnecessary residues does not**

MyoD1 WT



L122R (recognition) $\Delta\text{switch}=+9.17$ $\Delta\text{motif}=2.35$



A17R (unnecessary, imp-rank 50) $\Delta\text{switch}=+0.73$ $\Delta\text{motif}=0.27$



A42R (unnecessary, imp-rank 51) $\Delta\text{switch}=+2.88$ $\Delta\text{motif}=0.56$

